Weekly Work Report

DORA Ranking Portal – Caching Architecture & Institute Organogram Integration

Contributor: Lokesh-102214
Period: 19 August 2026 – 9 September 2026
Repository: `https://github.com/rishang19dx/dora-ranking-portal`
Source commit: 9ff3246 – *feat: integrate custom NIRF API caching and Institute Organogram* (parent 724b58c), committed 1 September 2026

Basis of this report

- **Scope.** Only work authored by Lokesh-102214 in commit 9ff3246 is covered. Other contributors' commits and later work are excluded.
- **Commit record.** The work is a single commit, 9ff3246, committed on 1 September 2026, with 15 files changed (+3,834 / -1 lines). The commit date is the author's statement; the commit page does not display a timestamp, so it has not been checked against git metadata. The reporting period (19 August–9 September 2026) is the author's stated timeline.
- **Hours are estimates.** Effort figures are the author's own build-time estimates (5 hours for the custom API, 2 hours for the organogram, 7 hours in total). No timesheet or time-tracking record exists for this period.
- **Line counts.** Exact counts are shown where the commit diff displays them. The other text files are reported as a combined total.
- **Creation status.** The diff shows `custom_api_nirf/app.py` as a new file (`@@ -0,0 +1,131 @@`), so this commit creates the file rather than modifying an earlier version.

Summary

Week	Dates & focus	Files changed	Est. effort (h)
1	19 Aug–9 Sep (commit dated 1 Sep): three-layer caching, organogram, integration documentation	15	7
Total		**15**	**7**

Lines added in this commit: 3,834. Lines removed: 1. Estimated effort for the period: 7 hours, matching the author's stated figures.

Week 1 (19 Aug–9 Sep): Caching, organogram and documentation

Estimated effort: 7 hours (estimate: 5 hours custom API, 2 hours organogram)

Custom API build (5 hours, estimate)

- **Backend API.** `custom_api_nirf/app.py` (131 lines) is a FastAPI wrapper that serves NIRF rankings from local JSON files by year and category. It supports filtering by state and city, a result limit, a second-generation IITs endpoint, and lookup by rank.
- **Layer 1, in-memory caching.** `load_data` is wrapped in `@lru_cache(maxsize=32)`, so each dataset is read from disk once. The cache is cleared when a scraper update succeeds.
- **Layer 2, frontend caching.** `src/utils/api.ts` provides `fetchNirfRankings`, which requests

data with a 24-hour revalidation window and the cache tag `nirf-data`.

- **Layer 3, webhook invalidation.** `src/app/api/revalidate/route.ts` handles revalidation requests. After a successful scrape, `/api/v1/scraper/update` posts the tag to this route, so the frontend drops stale data immediately. Failures in the webhook are logged and do not fail the scrape.
- **Data and scraper.** Six JSON datasets (2024 and 2025, covering engineering, innovation and overall rankings) are stored under `custom_api_nirf/data/`. `scraper.py` and `requirements.txt` support their generation and updates.

Institute organogram (2 hours, estimate)

- **Organogram chart.** `Institute_Organogram.md` (113 lines) is a Mermaid `graph LR` flowchart. It covers the executive level (ISC, Director, ExeCom, Audit, Recruitment cell), the Deans and Registrar with their sub-units, nine academic chairs or coordinators, and nine faculty bodies (seven school and two centre bodies).

Integration documentation

- **Integration guide.** `Integration_Docs.md` (60 lines) describes the three caching layers, the data storage layout, the organogram, a file tracking summary and the development time estimates.

Files changed

File	Status	Lines	Purpose
`Institute_Organogram.md`	Added	113	Mermaid organogram of the institute
`Integration_Docs.md`	Added	60	Integration and caching documentation
`custom_api_nirf/app.py`	Added	131	FastAPI wrapper, `lru_cache`, scraper webhook
`src/utils/api.ts`	Added	not shown	`fetchNirfRankings` with cache tags
`src/app/api/revalidate/route.ts`	Added	not shown	Revalidation webhook
`custom_api_nirf/scraper.py`	Added	not shown	NIRF data scraper
`custom_api_nirf/requirements.txt`	Added	not shown	Python dependencies
`package-lock.json`	Changed	not shown	Node dependency lock file
`custom_api_nirf/data/*.json` (6 files)	Added	not shown	NIRF 2024 and 2025 datasets
`custom_api_nirf/__pycache__/app.cpython-312.pyc`	Added	binary (6.28 KB)	Compiled bytecode, not source
Text files not itemised (11 files)		3,530	Combined total

Technical decisions

1. **Layered caching.** Memory caching on the backend, server-side caching on the frontend with a 24-hour window, and tag-based webhook invalidation after each scrape.
2. **Tag-based revalidation.** Revalidating by tag, rather than waiting for the time window to expire, keeps users from seeing stale rankings after a scrape.

3. **Organogram in Markdown.** The organogram is stored as Mermaid source in the repository, so it renders on GitHub and can be edited as text.

Open items noted in the commit

- The revalidation webhook URL is hard-coded to `http://localhost:3000/api/revalidate`. A deployed site would need this set through configuration.
- The compiled file `custom_api_nirf/__pycache__/app.cpython-312.pyc` is committed. Build artefacts of this kind are usually excluded with `.gitignore`.
- The commit does not record an end-to-end test of the webhook.